Chromatography Key

Three lookup tables. No abbreviations except in parentheses after the full name.

Biochemical Analysis — Birkbeck 2026

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1 Table 1: Every Chemical Mentioned in the Course

Organised by role. Functional groups listed for each compound so you can look them up in Table 2.

1.1 Elements and Ions (Ion Exchange Affinity Series)

Larger bare ions bind more strongly than smaller ones of the same charge because smaller ions have thicker hydration shells that cost more energy to strip.

Cation affinity series (weakest → strongest binding):

Ion	Charge	Context
Lithium (Li^+)	+1	Weakest-binding monovalent cation; thickest hydration shell.
Sodium (Na^+)	+1	Competing cation in ion exchange; NaCl gradient elution; mobile phase electrolyte.
Potassium (K^+)	+1	Buffer component (KH_2PO_4); intermediate monovalent affinity.
Ammonium (NH_4^+)	+1	Volatile buffer cation; MS-compatible.
Silver (Ag^+)	+1	Affinity series member; also used in silver staining.
Caesium (Cs^+)	+1	Strongest-binding monovalent cation; smallest hydration shell.
Magnesium (Mg^{2+})	+2	Divalent cation; stronger than any monovalent.
Zinc (Zn^{2+})	+2	Transition metal; affinity series member.
Cobalt (Co^{2+})	+2	Affinity series member.
Copper (Cu^{2+})	+2	Affinity series member.
Nickel (Ni^{2+})	+2	Affinity series member.
Calcium (Ca^{2+})	+2	Divalent cation; affinity series.
Lead (Pb^{2+})	+2	Affinity series member.
Barium (Ba^{2+})	+2	Strongest-binding divalent cation in series.

Anion affinity series (weakest → strongest binding):

Ion	Charge	Context
Fluoride (F^-)	−1	Weakest-binding monovalent anion.
Hydroxide (OH^-)	−1	NaOH mobile phase for anion chromatography (suppressed conductivity).
Acetate (CH_3COO^-)	−1	Volatile; MS-compatible buffer anion.
Formate ($HCOO^-$)	−1	Volatile; MS-compatible buffer anion.
Chloride (Cl^-)	−1	Ion chromatography analyte; NaCl competitor in ion exchange.
Bromide (Br^-)	−1	Ion chromatography analyte.
Nitrate (NO_3^-)	−1	Anion affinity series member.
Iodide (I^-)	−1	Anion affinity series; strong binding.
Thiocyanate (SCN^-)	−1	Strong-binding monovalent anion.
Oxalate ($C_2O_4^{2-}$)	−2	Divalent anion; stronger binding than monovalent.
Sulfate (SO_4^{2-})	−2	Divalent anion; strong binding.
Carbonate / Bicarbonate	−2/−1	Na_2CO_3/$NaHCO_3$ mobile phase for ion chromatography.
Phosphate (PO_4^{3-})	up to −3	Three pK_a values: 2.1, 7.2, 12.4. Non-volatile; incompatible with MS.
Citrate	up to −3	Trivalent anion; strongest binding in anion series.

1.2 Simple Molecules

Hydrogen chloride (HCl)

Permanent dipole (H–Cl bond). Dipole moment teaching example. Byproduct of silane reactions with silanol surface (trimethylchlorosilane end-capping).

Carbon dioxide (CO_2)

Linear, symmetric; no permanent dipole. Not detected by flame ionisation detector (FID) because it has no C–H bonds.

Methane (CH_4)

[EXCLUDED FROM EXAM] Boiling point −161 °C. No functional groups; only London dispersion forces. Chemical ionisation (CI) reagent gas in GC-MS.

Formaldehyde (methanal, $H_2C{=}O$)

Boiling point −19 °C. Carbonyl group gives dipole–dipole interactions (no H-bond donation). Silver stain developer component.

Carbon tetrachloride (CCl_4)

Symmetric tetrahedral geometry: four C–Cl dipoles cancel, net dipole zero. FID gives zero signal (no C–H bonds). Electron capture detector (ECD) is highly sensitive to it.

1.3 Solvents

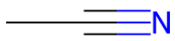
Water (H_2O)

Protic solvent. Hydrogen-bond donor and acceptor. Highly polar. Base solvent in reverse-phase HPLC. Volatile; compatible with ESI-MS.

Methanol (MeOH, CH_3OH)

Protic. Hydroxyl group. Boiling point +65 °C. Organic modifier in reverse-phase HPLC; weaker eluent than acetonitrile. Volatile; ESI-MS compatible.

Acetonitrile (MeCN, $CH_3C{\equiv}N$)



Aprotic. Nitrile group: H-bond acceptor only. Lower viscosity than methanol. Stronger eluent in reverse-phase HPLC. Preferred for LC-MS.

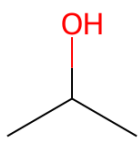
Ethanol

Protic solvent. Dielectric constant 24. Also used in Coomassie destaining solutions.

Propan-1-ol

Protic solvent. Mentioned in eluotropic series context for normal-phase HPLC.

Isopropanol (propan-2-ol, $(CH_3)_2CHOH$)



Protic. Hydroxyl group. Polar modifier in normal-phase HPLC: competes with analytes for silanol binding sites on silica.

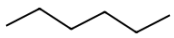
Acetone

Aprotic. Carbonyl group. Moderate polarity. Eluotropic series member; column cleaning solvent.

Pentane

Non-polar hydrocarbon. No functional groups. Discussed in hydrophobic effect context.

Hexane (C_6H_{14})



Non-polar hydrocarbon. No functional groups. Base solvent in normal-phase HPLC. The "weak" solvent in normal phase.

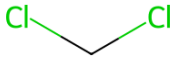
Isooctane (2,2,4-trimethylpentane)

Non-polar hydrocarbon. Eluotropic series base solvent for normal-phase HPLC.

Octanol

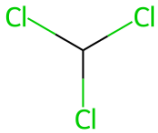
Long-chain alcohol. Partition coefficient reference solvent: log P = log(concentration in octanol / concentration in water).

Dichloromethane (DCM, CH_2Cl_2)



Two C–Cl bonds. Moderate polarity; the two C–Cl dipoles partially cancel. Normal-phase HPLC mobile phase component.

Chloroform ($CHCl_3$)



Three C–Cl bonds. Polar: net dipole does not cancel because of the hydrogen. More polar than DCM. Normal-phase HPLC component.

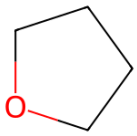
Diethyl ether

Aprotic. Ether oxygen: H-bond acceptor. Eluotropic series member for normal-phase HPLC.

Ethyl acetate

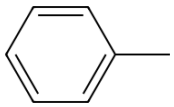
Ester group (H-bond acceptor). Eluotropic series member; extraction solvent.

Tetrahydrofuran (THF, C_4H_8O)



Aprotic. Cyclic ether oxygen: H-bond acceptor. Forms peroxides on air exposure. SEC mobile phase for synthetic polymers.

Toluene ($C_6H_5CH_3$)



Aromatic ring plus methyl group. Weakly polar via π-electrons. Normal-phase mobile phase component; also column test compound.

Dimethyl sulfoxide (DMSO)

Highly polarised S=O bond. Strong dipole moment. Dipole moment teaching example.

Pyridine

Aromatic ring with nitrogen lone pair. H-bond acceptor teaching example (lone pair on N not part of the aromatic π system). Silylation reaction medium.

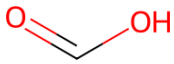
Glycerol

Three hydroxyl groups. Very polar, viscous. Loading buffer component for electrophoresis gels. Early matrix-assisted laser desorption ionisation (MALDI) matrix precursor.

1.4 Acids, Bases, and Buffer Components

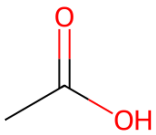
Trifluoroacetic acid (TFA, CF_3COOH)

Very strong acid: pK_a approximately 0.2. Three fluorines withdraw electron density from carboxyl group. Volatile but **suppresses ESI-MS signal** by ion-pairing. Replaced by formic acid in LC-MS work.



Formic acid (HCOOH)

Weak acid: pK_a approximately 3.7. One carboxyl group. Volatile. Standard acid additive for LC-ESI-MS at 0.1%.



Acetic acid (CH_3COOH)

Weak acid: pK_a approximately 4.8. Methyl plus carboxyl. Volatile. Reverse-phase HPLC pH control; component of ammonium acetate buffer.

Phosphoric acid (H_3PO_4)

Source of phosphate groups in nucleic acids. Three ionisable protons.

Methacrylic acid

Co-polymerised with styrene/divinylbenzene for weak acid cation exchange resin.

Sulfuric acid (H_2SO_4)

Strong acid. Regenerating solution for suppressor column in ion chromatography.

Ammonium acetate (NH_4CH_3COO)

Volatile buffer salt. Both ammonium (NH_4^+) and acetate (CH_3COO^-) evaporate. Buffering at pH 4.8 and 9.25. **The** MS-compatible buffer.

Ammonium formate (NH_4HCOO)

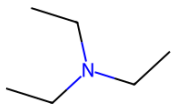
Volatile buffer salt. Alternative to ammonium acetate. MS-compatible.

Phosphate buffer (KH_2PO_4 / K_2HPO_4 / Na_2HPO_4)

Non-volatile. Three pK_a values: 2.1, 7.2, 12.4. **Incompatible with mass spectrometry**: phosphate deposits on internal MS surfaces.

Ammonia (NH_3, aqueous: NH_4OH)

Weak base. Conjugate acid NH_4^+ pK_a approximately 9.25. Volatile; MS-compatible.



Triethylamine (TEA, $(C_2H_5)_3N$)

Tertiary amine. Weak base. Competes with basic analytes for silanol sites, reducing tailing. Suppresses ESI-MS signal.

MES buffer (2-(N-morpholino)ethanesulfonic acid)

Zwitterionic biological buffer, pK_a 6.2. SDS-PAGE running buffer component.

MOPS buffer (3-(N-morpholino)propanesulfonic acid)

Zwitterionic biological buffer. Used in gel electrophoresis systems.

CAPS buffer (N-cyclohexyl-3-aminopropanesulfonic acid)

Zwitterionic biological buffer, pK_a 10.4. High-pH applications.

CHES buffer (N-cyclohexyl-2-aminoethanesulfonic acid)

Zwitterionic biological buffer.

Tris (tris(hydroxymethyl)aminomethane)

Amine buffer, pK_a 8.1. SDS-PAGE gel buffer: pH 6.8 (stacking gel), pH 8.8 (resolving gel). Three hydroxyl groups.

Borate / Boric acid

pK_a 9.2. Buffer for capillary electrophoresis (CE) and micellar electrokinetic chromatography (MEKC).

Bis-Tris

Gradient gel buffer used in Bolt gel systems. Lower pH range than Tris.

Bio-Lyte ampholytes

[EXCLUDED FROM EXAM] Carrier ampholytes (polyamine oligomers) for capillary isoelectric focusing (CIEF). Create pH gradient inside capillary.

Sodium carbonate (Na_2CO_3)

Ion chromatography eluant (paired with $NaHCO_3$). Also silver stain developer component.

Sodium chloride (NaCl)

Gradient elution salt in ion exchange chromatography. Na^+ or Cl^- competes for binding sites and displaces analytes.

1.5 Proteins and Enzymes

Trypsin

Serine protease. Cleaves C-terminal side of lysine (Lys, K) and arginine (Arg, R). Produces peptides that ionise well in ESI-MS (basic C-terminal residue).

Chymotrypsin

Serine protease. Cleaves after phenylalanine (Phe, F), tyrosine (Tyr, Y), tryptophan (Trp, W). Hydrophobic C-terminal residues.

Pepsin

Protease active at approximately pH 2. Used in hydrogen-deuterium exchange mass spectrometry (HDX-MS).

Lysozyme

Small protein (approximately 14.3 kDa). ESI-MS charge-state deconvolution example.

Insulin

Two chains linked by disulfide bonds. Electrophoresis example for DTT reduction.

Urease

Splits urea into ammonia and carbonate.

Erythropoietin (EPO)

Glycoprotein. Carbohydrate microheterogeneity detected by CE (multiple glycoforms).

Alpha-1-antitrypsin

Enzyme inhibitor. Travels from liver to lungs. Aggregation mutations cause disease.

Elastase

Protease controlled by alpha-1-antitrypsin.

Hemoglobin

[EXCLUDED FROM EXAM] pI 7.1–7.5. Molecular weight marker. CIEF example.

Myoglobin

pI 6.8–7.0. Molecular weight marker for SDS-PAGE.

Carbonic anhydrase

pI 6.0–6.5. Molecular weight marker (31 kDa).

Myosin

SDS-PAGE molecular weight marker (200 kDa).

Beta-galactosidase

Molecular weight marker (116 kDa).

Glycogen phosphorylase b

Molecular weight marker (97 kDa).

Bovine serum albumin (BSA)

Molecular weight marker (66.2 kDa). Also used as protein chiral stationary phase.

Ovalbumin

Molecular weight marker (45 kDa).

Soybean trypsin inhibitor

Molecular weight marker (21.5 kDa).

Aldolase

Practice mass drill protein (39.2 kDa subunit).

Alkaline phosphatase

Enzyme conjugate for Western blot detection (colorimetric).

Horseradish peroxidase (HRP)

Enzyme conjugate for Western blot chemiluminescence detection.

Transferrin

[EXCLUDED FROM EXAM] CIEF example protein.

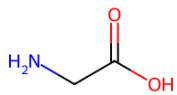
DNA polymerase

[EXCLUDED FROM EXAM] Sanger sequencing enzyme (capillary gel electrophoresis context).

1.6 Amino Acids

All amino acids share: α-amino group ($-NH_2$, pK_a approximately 9), α-carboxyl group ($-COOH$, pK_a approximately 2), and a variable side chain (R group). Residue mass = molecular weight − water (18 Da).

Glycine (Gly, G)



Residue mass 57 Da. Side chain: hydrogen only. Smallest amino acid. No functional group on side chain.

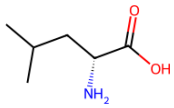
Alanine (Ala, A)

Residue mass 71 Da. Side chain: methyl group. Non-polar.

Valine (Val, V)

Residue mass 99 Da. Side chain: branched aliphatic (isopropyl). Non-polar.

Leucine (Leu, L)

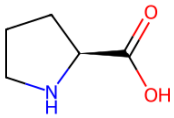


Residue mass 113 Da. Isobaric with isoleucine; cannot be distinguished by mass spectrometry alone. Hydrophobic branched aliphatic.

Isoleucine (Ile, I)

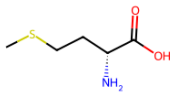
Residue mass 113 Da. Isobaric with leucine. Hydrophobic branched aliphatic.

Proline (Pro, P)



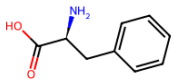
Residue mass 97 Da. Cyclic imino group. "Proline effect" in MS/MS: enhanced cleavage N-terminal to proline gives intense y ions.

Methionine (Met, M)



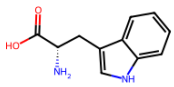
Residue mass 131 Da. Side chain: thioether (–S–CH_3). Non-polar. Cyanogen bromide cleavage site.

Phenylalanine (Phe, F)



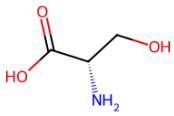
Residue mass 147 Da. Side chain: benzene ring. Non-polar, hydrophobic. UV-absorbing. Chymotrypsin cleavage site.

Tryptophan (Trp, W)



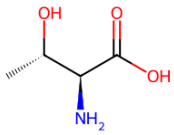
Residue mass 186 Da. Side chain: indole ring (bicyclic aromatic with N–H). Strong UV absorption at 280 nm. Chymotrypsin cleavage site. Immonium ion at 130.05 Da in MS/MS.

Serine (Ser, S)



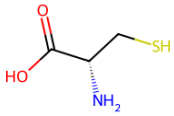
Residue mass 87 Da. Side chain: hydroxyl (–OH). Polar, uncharged. H-bond donor and acceptor.

Threonine (Thr, T)



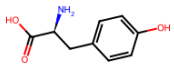
Residue mass 101 Da. Side chain: hydroxyl (–OH) on branched carbon. Polar, uncharged.

Cysteine (Cys, C)



Residue mass 103 Da. Side chain: thiol (–SH), pK_a approximately 8.3. Forms disulfide bonds. Reduce with DTT, alkylate with iodoacetamide (+57 Da).

Tyrosine (Tyr, Y)

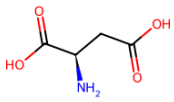


Residue mass 163 Da. Side chain: phenol (aromatic –OH), pK_a approximately 10.1. UV-absorbing. Chymotrypsin cleavage site.

Asparagine (Asn, N)

Residue mass 114 Da. Side chain: amide ($–CONH_2$). Polar, uncharged.

Aspartic acid (Asp, D)

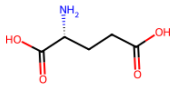


Residue mass 115 Da. Side chain: –COOH, pK_a approximately 3.9. Negatively charged at neutral pH.

Glutamine (Gln, Q)

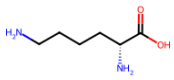
Residue mass 128 Da. Side chain: amide ($–CH_2–CONH_2$). Same nominal mass as lysine; distinguished only at high mass resolution.

Glutamic acid (Glu, E)



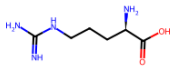
Residue mass 129 Da. Side chain: $–CH_2–COOH$, pK_a approximately 4.2. Negatively charged at neutral pH.

Lysine (Lys, K)



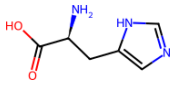
Residue mass 128 Da. Side chain: ε-amino group $(–CH_2)_4–NH_2$, pK_a approximately 10.5. Positively charged at physiological pH. Trypsin cleavage site. y_1 ion = 147.11 Da.

Arginine (Arg, R)



Residue mass 156 Da. Side chain: guanidinium group, pK_a approximately 12.5. Essentially always protonated. Trypsin cleavage site. y_1 ion = 175.12 Da.

Histidine (His, H)



Residue mass 137 Da. Side chain: imidazole ring, pK_a approximately 6.0. Partially protonated near neutral pH. pH-sensitive in ion exchange.

1.7 Nucleotides and Bases [EXCLUDED FROM EXAM – CGE section]

Adenine

[EXCLUDED FROM EXAM] Purine DNA base. Sanger sequencing context.

Guanine

[EXCLUDED FROM EXAM] Purine DNA base. Sanger sequencing context.

Cytosine

[EXCLUDED FROM EXAM] Pyrimidine DNA base. Sanger sequencing context.

Thymine

[EXCLUDED FROM EXAM] Pyrimidine DNA base (pairs with adenine). Sanger sequencing context.

Uracil

[EXCLUDED FROM EXAM] H-bond donor/acceptor teaching example: two C=O groups and two N–H groups.

dNTP (deoxyribonucleoside triphosphate)

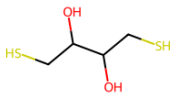
[EXCLUDED FROM EXAM] Normal sequencing substrate for DNA polymerase.

ddNTP (dideoxyribonucleoside triphosphate)

[EXCLUDED FROM EXAM] Chain-terminating sequencing reagent. Lacks 3'-OH, so no further extension.

1.8 Proteomics Reagents

Dithiothreitol (DTT)

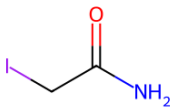


Two thiol groups (–SH), two hydroxyl groups (–OH). Reducing agent. Breaks disulfide bonds (–S–S–) to free thiols.

Beta-mercaptoethanol (2-mercaptoethanol)

Thiol reducing agent. Breaks disulfide bonds in SDS-PAGE sample preparation.

Iodoacetamide



Iodo group ($–CH_2I$) plus amide ($–CONH_2$). Alkylating agent. Reacts with free cysteine thiols: +57 Da carbamidomethyl modification.

Cyanogen bromide (BrCN)

Chemical cleavage agent. Cleaves peptide bonds C-terminal to methionine residues.

ICAT (Isotope-Coded Affinity Tag)

[EXCLUDED FROM EXAM] Quantitative proteomics labelling reagent.

iTRAQ (isobaric Tags for Relative and Absolute Quantitation)

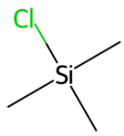
[EXCLUDED FROM EXAM] Isobaric labelling reagent for multiplexed quantitative proteomics.

Biotin

[EXCLUDED FROM EXAM] Affinity tag; captured by streptavidin beads.

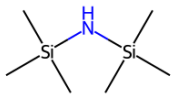
1.9 Derivatisation Reagents

Trimethylchlorosilane ($ClSi(CH_3)_3$)



Silicon with three methyl groups and one chlorine leaving group. End-capping reagent for HPLC silica. Also GC silylation reagent: replaces active hydrogens (–OH, –NH, –COOH) with $–Si(CH_3)_3$.

Hexamethyldisilazane (HMDS, $(CH_3)_3Si–NH–Si(CH_3)_3$)



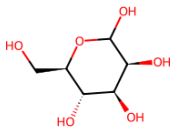
Alternative silylation reagent. Byproduct is ammonia (not HCl). Preferred because HCl causes side reactions.

Alkylchlorosilane (generic)

Reacts with Si–OH surface silanol groups to form bonded phases (C18, C8, phenyl, etc.) via Si–O–Si siloxane bond.

1.10 Carbohydrates

Glucose



Monosaccharide. Five hydroxyl groups, one ring oxygen. Very polar, no chromophore. Poorly ionisable in ESI-MS (forms sodium adducts). Detected by refractive index detector. Separated on aminopropyl phase. Component of cyclodextrin rings.

Galactose

Monosaccharide. Compared to 3,6-anhydrogalactose in agarose discussion: has more –OH groups, therefore more polar.

3,6-Anhydrogalactose

Component of agarose. Less polar than galactose because it has lost a water/hydroxyl group (intramolecular ether bridge).

Cyclodextrins (α, β, γ)

Rings of 6, 7, or 8 glucose units. Truncated cone: interior hydrophobic, exterior polar (multiple –OH). Chiral stationary phase: analyte enters cavity by hydrophobic inclusion; hydroxyl rim discriminates enantiomers.

Agarose

Polysaccharide of galactose and 3,6-anhydrogalactose. Neutral. Large-pore gel matrix for DNA electrophoresis (100 bp to >20 kb).

Dextran

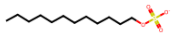
Polysaccharide. Base material for Sephadex SEC gel (crosslinked dextran).

Cellulose

Polysaccharide. Base for diethylaminoethyl cellulose (DEAE-cellulose) and carboxymethyl cellulose (CM-cellulose) ion exchangers. Also chiral phase material.

1.11 Surfactants and Gel/Electrophoresis Materials

Sodium dodecyl sulfate (SDS)

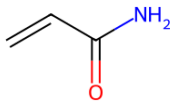


12-carbon alkyl chain plus sulfate head group (–O–SO_3^-, permanently charged). Anionic surfactant. Denatures proteins and coats them uniformly with negative charge (∼1.4 g SDS per gram protein) for SDS-PAGE.

Lithium dodecyl sulfate (LDS)

Variant of SDS in Bolt sample buffer. Same dodecyl sulfate anion, lithium counterion.

Acrylamide ($H_2C{=}CH{-}CO{-}NH_2$)



Vinyl group plus amide. Monomer for polyacrylamide gel. Pore size controlled by total acrylamide concentration. Neurotoxic as monomer.

Bisacrylamide (N,N'-methylenebisacrylamide)

Crosslinker for polyacrylamide gels. Bifunctional vinyl monomer connecting acrylamide chains.

TEMED (N,N,N',N'-tetramethylethylenediamine)

Polymerisation catalyst. Generates free radicals from ammonium persulfate to initiate acrylamide polymerisation.

Ammonium persulfate (APS)

Free radical initiator for gel polymerisation. Decomposes to sulfate radicals that start the chain reaction.

Nitrocellulose

Membrane for Western blot protein transfer from gel. Binds proteins non-specifically.

PVDF (polyvinylidene difluoride)

Alternative Western blot transfer membrane. More durable than nitrocellulose; requires methanol pre-wetting.

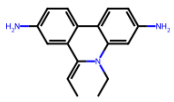
Polyimide

Outer protective coating on fused silica capillaries (GC and CE). Provides mechanical strength.

Coomassie Brilliant Blue

Aromatic dye with sulfonate groups. Protein stain: 0.1% in 50% methanol / 10% acetic acid. Binds hydrophobic regions and basic residues.

Ethidium bromide



Planar tricyclic aromatic with permanent positive charge (N^+). Intercalates between DNA base pairs. Fluorescence enhanced 20–30 fold when intercalated. UV visualisation of DNA in agarose gels.

SYBR Green, SYBR Gold

Safer DNA intercalating dyes. Alternatives to ethidium bromide.

Luminol

Chemiluminescence substrate for horseradish peroxidase (HRP) Western blot detection. Emits light when oxidised.

Hydrogen peroxide (H_2O_2)

Co-substrate with luminol in chemiluminescence Western blot detection.

Silver nitrate ($AgNO_3$)

Silver staining reagent. Ag^+ binds proteins in gel, then is reduced to metallic silver by formaldehyde/Na_2CO_3 developer.

Methylcellulose

[EXCLUDED FROM EXAM] Linear polymer for replaceable capillary gel electrophoresis (CGE) matrix.

Ampholytes (carrier ampholytes)

[EXCLUDED FROM EXAM] Polyamine oligomers for isoelectric focusing (IEF) pH gradient generation.

IPG strips (immobilised pH gradient)

[EXCLUDED FROM EXAM] Immobilised pH gradient strips for two-dimensional gel electrophoresis.

1.12 Size Exclusion Chromatography (SEC) Gels

Sephadex

Crosslinked dextran gel beads for SEC. Aqueous mobile phase. Separates by molecular size.

Sepharose

Crosslinked agarose gel beads for SEC. Large pore sizes for high-molecular-weight proteins.

BioGel

Crosslinked polyacrylamide beads for SEC.

1.13 Silica and Column Materials

Fused silica / fused quartz

Capillary material for GC and CE. Amorphous SiO_2. Surface silanol groups (Si–OH) drive bonded phase chemistry and electroosmotic flow.

Alumina (Al_2O_3)

Alternative normal-phase HPLC stationary phase. More basic surface than silica.

Sodium silicate

Precursor for angular (irregular) silica particle synthesis.

Tetraethoxysilane (TEOS)

Precursor for spherical silica synthesis. Hydrolysed with NaOH to produce uniform spherical particles (sol-gel process).

Polystyrene (crosslinked with divinylbenzene)

Ion exchange resin backbone polymer. Hydrophobic; provides the mechanical structure for sulfonate or quaternary ammonium functional groups.

1.14 HPLC Bonded Phases

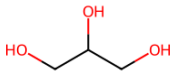
C18 (octadecylsilane)

18-carbon hydrocarbon chain bonded to silica. Most common reverse-phase stationary phase. Separates by hydrophobicity.

Phenyl bonded phase

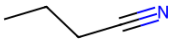
Aromatic ring bonded to silica. Provides π–π selectivity for aromatic analytes, different from C18.

Diol bonded phase



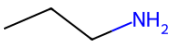
Two adjacent hydroxyl groups. Polar bonded phase for normal-phase HPLC. H-bond donor and acceptor.

Cyano (nitrile) bonded phase



–C≡N group bonded to silica. Intermediate polarity; can operate in normal-phase or reverse-phase mode.

Aminopropyl bonded phase

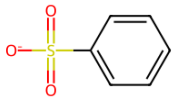


–NH_2 group on propyl spacer. Used for sugar separations (H-bonding with hydroxyl groups). Also weak anion

exchanger at low pH.

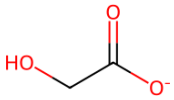
1.15 Ion Exchange Phases

Sulfonate phase (strong cation exchanger, SCX)



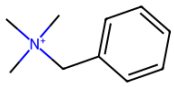
–SO_3^-: permanent negative charge at all pH values. Binds cations.

Carboxymethyl cellulose (CM-cellulose, weak cation exchanger, WCX)



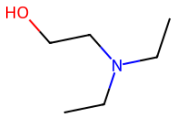
–CH_2–COO^-: negatively charged above pK_a approximately 4. Loses charge at very low pH.

Quaternary ammonium phase (strong anion exchanger, SAX)



–$N^+(CH_3)_3$: permanent positive charge at all pH values. Binds anions.

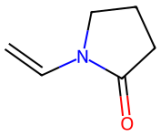
Diethylaminoethyl cellulose (DEAE-cellulose, weak anion exchanger, WAX)



–$N^+H(C_2H_5)_2$: positively charged below pK_a approximately 11. Loses charge at very high pH.

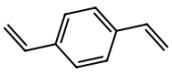
1.16 Solid-Phase Extraction (SPE) Sorbent Monomers

N-vinylpyrrolidone (NVP)



Hydrophilic monomer of hydrophilic-lipophilic balance (HLB) sorbent. Lactam ring: H-bond acceptor.

Divinylbenzene (DVB)



Lipophilic crosslinker monomer of HLB sorbent. Aromatic ring with two vinyl groups.

Accell Plus CM
Weak cation exchanger SPE sorbent. Carboxymethyl groups.

Accell Plus QMA
Strong anion exchanger SPE sorbent. Quaternary methylammonium groups.

1.17 GC Stationary Phases

Polydimethylsiloxane (SE-30, OV-101)
Non-polar GC stationary phase. Usable 50–350 °C. Separates by boiling point (volatility) only.

OV-5 (5% phenyl dimethylsiloxane)
Slightly polar GC phase. Small phenyl content gives weak π-interaction selectivity.

OV-17 (50% phenyl dimethylsiloxane)

Moderately polar GC phase. Higher phenyl content for aromatic selectivity.

Carbowax 20M (polyethylene glycol, PEG)

Polar GC stationary phase, 0–250 °C. Ether oxygens act as H-bond acceptors. Best for oxygenated compounds.

Squalane

Non-polar GC phase for saturated hydrocarbons. Branched C_{30} alkane.

Apiezon L

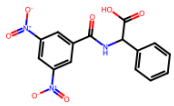
Branched-chain alkane, non-polar, 60–325 °C. High-temperature non-polar phase.

Dexil 300

Carborane dimethylsilicone. High temperature stability GC phase.

1.18 Chiral Selectors

Pirkle phase (dinitrobenzoyl phenylglycine)



Three-point interaction: π–π stacking (electron-poor dinitrophenyl ring), H-bonding (amide), steric fit at chiral centre. One enantiomer achieves all three contacts; the other cannot.

Naphthylleucine

Pirkle-type chiral stationary phase. Naphthyl ring provides π–π stacking. From 2025 exam question 3b.

Crown ether

Cyclic polyether. Chiral variants used for CE enantioseparation. Selective inclusion of ammonium groups.

Cholic acid

Steroid-based bile salt. Used in MEKC chiral separation. Critical micelle concentration (CMC) approximately 14 mM.

1.19 CE Additives

CTAB (cetyltrimethylammonium bromide)

Cationic surfactant. Reverses electroosmotic flow (EOF) direction by coating capillary wall with positive charges. CMC 1.3 mM.

Polybrene (hexadimethrine bromide)

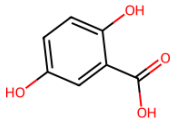
Cationic polymer coating. Reverses EOF and prevents protein adsorption to capillary wall.

DTAB (dodecyltrimethylammonium bromide)

[EXCLUDED FROM EXAM] Cationic surfactant. CMC 14 mM. MEKC context.

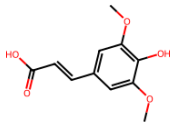
1.20 MALDI Matrices

2,5-Dihydroxybenzoic acid (DHB)



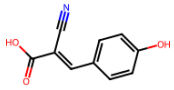
Two phenolic –OH plus one –COOH. UV-absorbing. Proton donor during laser desorption. Common matrix for peptides and proteins.

Sinapinic acid (3,5-dimethoxy-4-hydroxycinnamic acid)



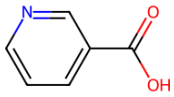
Two methoxy groups, one phenolic –OH, one conjugated –COOH. Strong UV absorption. Matrix for large proteins (>10 kDa).

α-Cyano-4-hydroxycinnamic acid (CHCA)



Nitrile (–C≡N), phenolic –OH, conjugated –COOH. Very strong UV absorption. Matrix for small peptides and small molecules.

Nicotinic acid (niacin)



Pyridine ring nitrogen plus –COOH. UV-absorbing. One of the first MALDI matrices.

1.21 Drugs and Named Analytes

Caffeine

HPLC practical analyte (quantification in beverages). UV-active purine derivative.

Paracetamol (acetaminophen)

Phenol group (acidic –OH). Amide group. Partition coefficient calculation example.

Ibuprofen

Chiral compound. pK_a 4.9. Carboxylic acid group. Extraction calculation example (log P and recovery).

Rapamycin

Complex macrolide. UV-absorbing. Diode array detector (DAD) story: same retention time as another compound, but different UV spectrum revealed it was not rapamycin.

Cocaine

Detected on currency by direct analysis in real time mass spectrometry (DART-MS). Practice problem: forensic extraction, pK_a 8.7, log P 2.30.

Naproxen

Chiral nonsteroidal anti-inflammatory drug (NSAID). Naphthyl ring stacks with Pirkle phase selector via π–π interaction.

Ascorbic acid (vitamin C)

Electrochemically active HPLC analyte. Multiple hydroxyl groups.

1.22 Aromatic Test Compounds

Benzene

Dipole moment 0 D (symmetric). GC elution order: boiling point 80 °C, elutes first on non-polar column.

Chlorobenzene

GC elution order: boiling point 132 °C. One C–Cl bond.

1,3-Dichlorobenzene

GC elution order: boiling point 172 °C. Two C–Cl bonds.

1,3,5-Trichlorobenzene

GC elution order: boiling point 208 °C, elutes last on non-polar column. Three C–Cl bonds.

Benzamide

HPLC test analyte on C4/C8/C18 columns. Amide group: polar, H-bond donor and acceptor.

Propylbenzene

HPLC test analyte. Aromatic ring plus propyl chain; moderate hydrophobicity.

Butylbenzene

HPLC test analyte. Aromatic ring plus butyl chain; more hydrophobic than propylbenzene.

Dimethyl phthalate

HPLC test analyte. Two ester groups on aromatic ring.

Biphenyl

HPLC test analyte. Two linked aromatic rings; hydrophobic, strong UV absorption.

Styrene

Non-polar: vinyl group plus aromatic ring. No ionisable groups. Monomer for polystyrene resins.

Mandelic acid

Polar: two hydroxyl groups plus carboxylic acid. 2025 exam compound.

Cyclohexanol

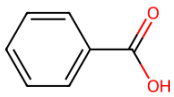
One hydroxyl group. Moderate polarity. 2025 exam compound.

1.23 Other Named Analytes and Teaching Examples

Diallyl disulfide

Two allyl groups linked by –S–S–. Volatile sulfur compound from garlic. Headspace GC; detected by ECD.

Benzoic acid esters (methyl, ethyl, propyl, butyl)



Aromatic ring plus ester (–COO–R). Increasingly hydrophobic with longer alkyl chain. Reverse-phase HPLC test compounds: longer chain = longer retention on C18.

Metalloporphyrin

Dipole–ion interaction teaching example. Metal ion (M^+) coordinated to nitrogen and oxygen atoms.

Tetramethylammonium

Quaternary ammonium, permanent positive charge. H-bond teaching example: cannot donate or accept H-bonds despite having nitrogen.

Ethyl formate

Ester group. H-bond acceptor teaching example (C=O lone pairs accept, but no H to donate).

1.24 Isotopes

Neon-20, Neon-22

Thomson's mass spectrometry experiment: two isotopes separated by deflection in electric and magnetic fields,

proving isotopes exist.

Carbon-12

Reference isotope for atomic mass scale (defined as exactly 12 u).

Carbon-13

Natural abundance 1.1%. Causes M+1 isotope peak in mass spectra; intensity proportional to number of carbons.

Nickel-63

Radioactive β-emitter source in the electron capture detector (ECD). Provides standing current of thermal electrons.

Tritium (^{3}H)

Alternative radioactive source for ECD.

2 Table 2: Functional Group Lookup

For each functional group: its chemical properties, then one sentence per separation technique describing its effect and why.

Hydroxyl (–OH)

Polar. Hydrogen-bond donor and acceptor. pK_a approximately 15 (aliphatic alcohol) or approximately 10 (phenol). Not ionised at normal working pH (except phenols above pH 10).

• Reduces retention in reverse-phase HPLC on C18 (octadecyl) stationary phase because the hydroxyl group hydrogen-bonds preferentially with the aqueous mobile phase rather than partitioning into the non-polar hydrocarbon chain layer.

• Increases retention in normal-phase HPLC on bare silica because the hydroxyl hydrogen-bonds to the surface silanol (Si–OH) groups, holding the analyte on the stationary phase.

• Raises boiling point in gas chromatography because intermolecular hydrogen bonds between hydroxyl groups hold molecules in the liquid phase; silylation replaces –OH with –O–Si$(CH_3)_3$, which cannot hydrogen-bond, lowering the boiling point and making the compound volatile enough for GC analysis.

• No direct role in ion exchange chromatography because the hydroxyl group is not ionised at working pH (pK_a approximately 15).

• Not detected by UV absorption unless attached to an aromatic ring with conjugation; detected by refractive index detector (universal).

Carboxylic Acid (–COOH / –COO^-)

Acidic. pK_a approximately 2–5. Below pK_a: neutral, protonated (–COOH). Above pK_a: ionised, deprotonated (–COO^-, negative charge). Hydrogen-bond donor (when protonated) and acceptor.

• At pH above pK_a, retention in reverse-phase HPLC on C18 drops sharply because the ionised carboxylate (–COO^-) is fully solvated by water and has no affinity for the non-polar stationary phase; at pH below pK_a, the neutral acid (–COOH) retains some hydrophobic contact through the rest of the molecule and is retained longer.

• At pH near the pK_a, the peak broadens severely or disappears in reverse-phase HPLC because half the molecules are neutral (slow) and half are ionised (fast), and they interconvert continuously during transit through the column.

• When ionised (–COO^-), binds to anion exchange surfaces (positively charged stationary phases such as diethylaminoethyl cellulose, DEAE-cellulose, or quaternary ammonium resins) via electrostatic attraction.

• Retained by mixed-mode anion exchange solid-phase extraction sorbent (Oasis MAX) because the quaternary ammonium group on the sorbent attracts the anionic carboxylate.

• Raises boiling point in gas chromatography because the carboxylic acid hydrogen-bonds intermolecularly; silylation converts –COOH to –COO–Si$(CH_3)_3$, blocking the hydrogen bond and lowering the boiling point.

• Ionisable in electrospray ionisation mass spectrometry: detected in negative ion mode as $[M–H]^-$ when the carboxyl proton is lost.

Amine (–NH_2, –NHR, –NR_2)

Basic. pK_a approximately 9–11 (aliphatic), approximately 4.6 (aniline). Above pK_a: neutral (–NH_2). Below pK_a: protonated (–NH_3^+, positive charge). Hydrogen-bond donor (when N–H present) and acceptor.

• At pH below pK_a, retention in reverse-phase HPLC on C18 drops because the protonated amine (–NH_3^+) is charged and hydrophilic, preferring the aqueous mobile phase over the non-polar stationary phase.

• On reverse-phase HPLC columns with residual silanol groups (not end-capped), the protonated amine (–NH_3^+) binds ionically to deprotonated silanols (Si–O^-), producing peak tailing because this additional strong interaction occurs at random surface sites.

• When protonated (–NH_3^+), binds to cation exchange surfaces (negatively charged stationary phases such as carboxymethyl cellulose, CM-cellulose, or sulfonate resins) via electrostatic attraction.

• Retained by mixed-mode cation exchange solid-phase extraction sorbent (Oasis MCX) because the sulfonate group on the sorbent attracts the cationic amine.

• In capillary zone electrophoresis, a protonated amine gives the analyte a positive charge, adding to its electrophoretic mobility in the same direction as electroosmotic flow, so it migrates faster and arrives at the detector before neutral molecules.

• Ionisable in electrospray ionisation mass spectrometry: detected in positive ion mode as $[M+H]^+$ when the amine accepts a proton.

Sulfonate ($-SO_3^-$)

Very strong acid: pK_a approximately 0. Always deprotonated. Permanent negative charge at any working pH.

• Provides essentially no retention in reverse-phase HPLC on C18 because the permanent negative charge makes the molecule extremely hydrophilic; the charged sulfonate is fully solvated by water and does not interact with the hydrocarbon stationary phase.

• Used as the functional group on strong cation exchange resins (polystyrene-divinylbenzene or silica-based); the permanent negative charge binds positively charged analytes (cations) regardless of pH.

• Used as the functional group on the mixed-mode cation exchange solid-phase extraction sorbent (Oasis MCX), providing cation-binding capacity in addition to the hydrophobic character of the polymer backbone.

• In capillary zone electrophoresis, a sulfonate gives the analyte a permanent negative charge, opposing electroosmotic flow; the analyte migrates more slowly than neutrals and arrives at the detector later.

Quaternary Ammonium ($-N^+R_3$)

Permanently positive charge. No proton to lose. Charge independent of pH.

• Provides essentially no retention in reverse-phase HPLC on C18 because the permanent positive charge makes the molecule extremely hydrophilic.

• Used as the functional group on strong anion exchange resins; the permanent positive charge binds negatively charged analytes (anions) regardless of pH.

• Used as the functional group on the mixed-mode anion exchange solid-phase extraction sorbent (Oasis MAX), providing anion-binding capacity in addition to the hydrophobic polymer backbone.

Silanol (Si–OH)

Weakly acidic: pK_a approximately 4–5. At pH above 4–5, deprotonates to $Si–O^-$ (negative charge).

• In normal-phase HPLC, the silanol groups on the bare silica surface are the active sites that retain polar analytes via hydrogen bonding; more polar analytes interact more strongly with these groups and elute later.

• In reverse-phase HPLC, residual uncapped silanols cause peak tailing for basic analytes because the deprotonated silanol ($Si–O^-$) forms an ionic bond with protonated amines ($-NH_3^+$); end-capping with trimethylsilyl groups blocks these sites.

• In capillary zone electrophoresis, deprotonated silanols ($Si–O^-$) on the capillary inner wall create the negative surface charge that drives electroosmotic flow; more deprotonation at higher pH produces stronger flow.

• The silanol is also the reactive site for all bonded-phase synthesis: the Si–OH reacts with alkylchlorosilane reagents to form Si–O–Si siloxane bonds, attaching C18 chains, phenyl groups, or other modifiers to the surface.

Carbonyl (C=O)

Polar. Hydrogen-bond acceptor only (no hydrogen to donate). Found in ketones, aldehydes, amides, esters.

• Reduces retention slightly in reverse-phase HPLC on C18 because the polar C=O group prefers the aqueous phase, but less dramatically than –OH because it cannot donate hydrogen bonds (only accept them).

• Increases retention in normal-phase HPLC on bare silica via hydrogen bonding between the carbonyl oxygen (acceptor) and the silanol hydrogen (donor).

• Raises boiling point in gas chromatography less than –OH does, because the carbonyl enables dipole-dipole interactions but not the stronger O–H···O hydrogen bonds (compare formaldehyde at −19 °C with methanol at +65 °C — the difference is hydrogen-bond *donation*).

Amide (–CO–NH–)

The peptide bond. C=O is hydrogen-bond acceptor; N–H is hydrogen-bond donor. Partially rigid due to resonance (C–N has partial double-bond character).

• In tandem mass spectrometry (MS/MS) with collision-induced dissociation (CID), the amide bond is the weakest bond in the peptide backbone and fragments preferentially, producing b ions (N-terminal fragment, charge retained on oxazolone ring) and y ions (C-terminal fragment, charge on protonated N-terminus, mass includes water).

• In chiral HPLC on Pirkle-type phases, the amide groups on both the analyte and the chiral selector form hydrogen bonds that contribute one of the three required interaction points for enantioseparation.

Aromatic Ring (Benzene, Phenyl, Naphthyl, Indole)

Planar. Hydrophobic. π-electron system. UV-absorbing.

• Increases retention in reverse-phase HPLC on C18 because the hydrophobic flat ring surface partitions into the

non-polar stationary phase layer.

• On a phenyl bonded-phase column, the analyte aromatic ring can stack face-to-face with the column's phenyl group via π–π interactions, giving different selectivity from C18 for mixtures of aromatic versus aliphatic compounds.

• In chiral HPLC on Pirkle-type phases, an electron-rich aromatic ring on the analyte stacks with the electron-poor 3,5-dinitrophenyl ring on the chiral selector (the nitro groups withdraw electron density, making the selector ring a π-acid), providing one of the three required interaction points.

• Detected by UV absorption because the π-electrons absorb ultraviolet light; polycyclic aromatics (naphthalene, anthracene) also fluoresce, enabling fluorescence detection at 10–1000 times greater sensitivity.

• Not readily ionisable in electrospray ionisation mass spectrometry unless the aromatic ring bears an ionisable substituent (–OH, –COOH, $–NH_2$); a pure hydrocarbon aromatic (e.g., styrene, anthracene) is difficult to ionise by ESI.

Nitrile (–C≡N)

Polar dipole. Hydrogen-bond acceptor only (nitrogen lone pair). Intermediate polarity between –OH and alkyl.

• On a nitrile (cyano) bonded-phase HPLC column, provides selectivity intermediate between bare silica (normal phase) and C18 (reverse phase); can be operated in either mode depending on whether the mobile phase is non-polar (normal phase behaviour) or aqueous (reverse phase behaviour).

Phenol (Aromatic –OH)

More acidic than aliphatic –OH: pK_a approximately 10. Below pK_a: neutral. Above pK_a: deprotonated ($Ar–O^-$, negative charge).

• In reverse-phase HPLC on C18, retention drops sharply above pH 10 because the deprotonated phenolate ($Ar–O^-$) is charged and hydrophilic; keep pH below 10 to maintain the neutral form and normal retention.

• UV-absorbing because the aromatic ring is a chromophore; ionisable in electrospray ionisation mass spectrometry because the phenolic proton can be lost (negative mode).

Thiol (–SH)

Weakly polar. Reducing agent. pK_a approximately 8–9.

• In proteomics sample preparation, free thiols on cysteine residues must be reduced with dithiothreitol (DTT) and then alkylated with iodoacetamide (+57 Da per cysteine) before mass spectrometry; otherwise, random disulfide bonds (–S–S–) form between different cysteines, giving unpredictable peptide masses.

Halogen (–F, –Cl, –Br)

Electronegative atoms. C–X bonds are polar (dipole). Halogens capture thermal electrons.

• In gas chromatography with an electron capture detector (ECD), halogenated compounds are detected with extreme sensitivity (detection limit 10^{-14}) because the halogen atoms capture β-electrons emitted by the nickel-63 source, reducing the standing current between the detector electrodes.

• More halogens on a molecule increase its molecular weight, boiling point, and retention on gas chromatography columns; adding chlorine substituents to benzene (chlorobenzene → dichlorobenzene → trichlorobenzene) increases retention on a non-polar column because boiling point rises.

• Not detected by flame ionisation detector (FID) if the molecule has no C–H bonds (e.g., carbon tetrachloride CCl_4 gives zero FID signal because there is nothing to burn).

3 Table 3: Technique Lookup

For each separation technique: what the stationary phase is, what the mobile phase is, what molecular property drives separation, what gradient is used, and which detectors work.

Reverse-Phase HPLC (RP-HPLC)

Stationary phaseSilica bonded with C18 (octadecyl), C8, C4 hydrocarbon chains, or phenyl groups, via siloxane (Si–O–Si) linkage. Mobile phaseWater (weak solvent) mixed with methanol or acetonitrile (strong solvent). pH adjusted with formic acid, acetic acid, or ammonium acetate buffer. Separation propertyHydrophobicity. Non-polar analytes partition into the hydrocarbon layer and are retained; polar analytes stay in the aqueous phase and elute first. GradientIncrease organic solvent proportion to elute increasingly hydrophobic analytes. pH effectIonising a functional group ($–COOH \rightarrow –COO^-$ or $–NH_2 \rightarrow –NH_3^+$) makes the analyte more polar, reducing retention. Never run at pH = pK_a. DetectorsUV/visible (needs chromophore), ESI-MS (needs ionisable group), refractive index (universal, low sensitivity), fluorescence (needs rigid conjugated system).

Normal-Phase HPLC (NP-HPLC)

Stationary phaseBare silica (Si–OH silanol groups) or polar bonded phases (diol, cyano, aminopropyl). Mobile phaseNon-polar base solvent (hexane) with polar modifier (isopropanol, ethyl acetate). Separation propertyPolarity. Polar analytes hydrogen-bond to the silanol surface and are retained; non-polar analytes have weak interaction and elute first. GradientIncrease polar modifier proportion. Cannot easily do gradient (slow equilibration between solvents and adsorbed layer). DetectorsUV/visible, refractive index. Not typically coupled to ESI-MS (non-aqueous solvents).

Ion Exchange Chromatography

Stationary phaseNegatively charged surface = cation exchanger (sulfonate $–SO_3^-$ or carboxymethyl $–COO^-$). Positively charged surface = anion exchanger (quaternary ammonium $–N^+R_3$ or diethylaminoethyl $–NH^+Et_2$). On polystyrene-divinylbenzene (small ions) or cellulose (proteins — needs large pores). Mobile phaseAqueous buffer containing a competing electrolyte (sodium chloride). Separation propertyCharge. Analyte ions of opposite charge to the surface bind; stronger charge or higher charge density = stronger binding = later elution. GradientIncrease ionic strength (add more salt). Na^+ or Cl^- competes for binding sites and displaces analytes. Protein ruleIf buffer pH < protein isoelectric point (pI): protein is net positive → use cation exchanger. If pH > pI: net negative → use anion exchanger. Small ion orderLarger bare ions bind more strongly than smaller ones of the same charge because smaller ions have thicker hydration shells that cost more energy to strip ($Li^+ < Na^+ < K^+ < Cs^+$). DetectorsUV/visible (proteins, organic ions). Suppressed conductivity (simple inorganic ions: suppressor column removes mobile phase ions, leaving analyte ions detectable against water baseline).

Size Exclusion Chromatography (SEC)

Stationary phasePorous gel. Crosslinked dextran/agarose (proteins, aqueous mobile phase) or crosslinked polystyrene (synthetic polymers, organic solvent). Must have negligible chemical interaction with analytes. Mobile phaseConstant composition (isocratic). Aqueous buffer for proteins. No gradient. Separation propertyMolecular size. Large molecules are excluded from pores and travel only through the void volume → elute first. Small molecules enter pores and explore a larger volume → elute last. Opposite to all other modes. CalibrationLog(molecular weight) versus elution volume is approximately linear between exclusion limit and total permeation limit. DetectorsRefractive index (universal). UV/visible if analytes absorb.

Gas Chromatography (GC)

Stationary phaseNon-polar: polydimethylsiloxane (SE-30). Polar: polyethylene glycol (Carbowax). Capillary columns (fused silica, 5–100 m, 0.1–0.7 mm inner diameter) have far higher resolution than packed columns. Mobile phaseInert carrier gas. Nitrogen (best efficiency at low flow, packed columns), helium (good efficiency at higher flow, capillary), hydrogen (best tolerance of high flow, fastest analysis). Separation propertyVolatility (boiling point) on non-polar phases. On polar phases, hydrogen bonding and dipole interactions additionally retain polar analytes, altering elution order versus non-polar phase. GradientTemperature programming: increase column temperature during the run. Early-eluting compounds resolve at low temperature; late-eluting compounds sharpen at higher temperature. DerivatisationSilylation replaces active hydrogens (–OH, –NH, –COOH) with $–Si(CH_3)_3$, blocking hydrogen bonding and lowering boiling point. Internal standardAlways required for quantitation because injection volume is imprecise (especially split injection on capillary columns). DetectorsFlame ionisation detector (FID): responds to all compounds with C–H bonds, destructive. Thermal conductivity detector (TCD): universal, low sensitivity, non-destructive. Electron capture detector (ECD): extremely sensitive to halogens, nitro groups, sulfur; uses nickel-63 source.

Capillary Zone Electrophoresis (CZE)

Stationary phaseNone (free solution in fused-silica capillary, 50 μm inner diameter). Mobile phaseBuffer solution filling the capillary. pH controls both analyte charge and electroosmotic flow strength. Separation propertyCharge-to-size ratio. Electrophoretic mobility $\mu = q/6\pi\eta r$. Higher charge relative to size = faster migration. Electroosmotic flow (EOF)Silanol groups on capillary wall deprotonate to Si–O$^-$ at pH > 3 → cation layer near wall → applied voltage drags cations to cathode → bulk solution follows. Flat plug flow profile (no broadening from laminar flow). Typically stronger than individual ion mobility, so everything reaches detector, even anions. Migration orderSmall highly-charged cations (fastest) → large weakly-charged cations → neutrals (EOF only) → large weakly-charged anions → small highly-charged anions (slowest). pH effectHigher pH → more Si–O$^-$ → stronger EOF → all peaks faster but possibly less resolved. Also changes analyte ionisation state. Temperature effectHigher temperature → lower buffer viscosity → all ions faster → shorter migration times, possibly less resolution. DetectorsUV absorption (on-capillary detection cell). Conductivity. Mass spectrometry (CE-MS).

Solid-Phase Extraction (SPE)

Sorbent typesHLB (hydrophilic-lipophilic balance): copolymer of N-vinylpyrrolidone (polar) and divinylbenzene (non-polar); catches both polar and non-polar analytes. MAX: HLB plus quaternary ammonium; binds anions. MCX: HLB plus sulfonate; binds cations. C18 cartridge: pure reverse-phase. WorkflowCondition cartridge → load sample → wash (remove matrix, keep analyte) → elute (release analyte with strong solvent). Selection ruleAnalyte has –COOH or –SO_3H (anionic) → MAX. Analyte has –NH_2 (cationic when protonated) → MCX. Unsure or neutral analyte → HLB.

Chiral HPLC

Stationary phasePirkle phase: amino acid derivative (e.g., 3,5-dinitrobenzoyl phenylglycine) bonded to silica. Cyclodextrin: glucose ring bonded to silica. ProblemEnantiomers have identical physical properties in a symmetric environment; they co-elute on achiral columns. MechanismPirkle: three-point interaction (π–π stacking, hydrogen bonding, steric fit at chiral centre). One enantiomer achieves all three contacts; the other cannot. Cyclodextrin: hydrophobic inclusion complex in cavity; chiral –OH rim discriminates enantiomers via different hydrogen-bonding geometries. AlternativeDiastereoisomer derivatisation: react both enantiomers with a single-enantiomer chiral reagent; products have different physical properties and separate on a standard achiral column.

End of key. v1 draft.